Arista Access Node HEB-ACC-25 Hardware/Environmental Fault SOP (ALARM_4855)

Node: HEB-ACC-25 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_4855

Description

This SOP outlines the procedure to investigate and resolve a critical hardware/environmental fault on the Arista Access Node HEB-ACC-25 at the Hebbal Edge site. The fault is characterized by ASIC memory parity errors, line card crashes, and power supply redundancy loss. This failure domain occurs due to the inherent design of the Arista 7250X-64 switch, which uses a distributed architecture with multiple ASICs and line cards to provide high-performance switching and routing capabilities. In the event of a hardware failure, the ASICs and line cards may experience memory parity errors, leading to crashes and subsequent loss of power supply redundancy. The critical alarm code ALARM_4855 is generated when the system detects a hardware failure that impacts the switch's ability to forward traffic. This SOP is designed to identify the root cause of the failure, isolate the affected component, and implement a resolution to restore normal switch operation.

Pre-requisite Checks

- Verify that the remote hands are dispatched to the Hebbal Edge site to provide physical access to the affected switch.

- Check the status of the redundant links connected to the switch to ensure they are operational and not impacted by the fault.

- Review the switch's configuration and log files to identify any recent changes or issues that may have contributed to the failure.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

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Command: show system internal memory, Expected Output: Memory parity errors detected, Description: This command displays the system's internal memory information, including any parity errors that may have occurred. The output will help identify if the ASIC memory parity error is the root cause of the failure.

Command: show interface status, Expected Output: Line card status: Down, Description: This command displays the status of the line cards connected to the switch. If a line card is down, it may indicate a
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crash or hardware failure.

Command: show power supply status, Expected Output: Power supply redundancy: Lost, Description: This command displays the status of the power supplies connected to the switch. If power supply redundancy is lost, it may indicate a hardware failure.
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Resolution Steps

Step 1: Step: Power cycle the switch, Command: reload, Description: This step involves powering off the switch and then restarting it to clear any transient issues. The reload command will initiate the power cycle process.

Step 2: Step: Replace the faulty ASIC or line card, Command: replace slot , Description: This step involves replacing the faulty ASIC or line card with a new one. The replace command will initiate the replacement process and ensure that the switch is brought online with the new component.

Step 3: Step: Verify power supply redundancy, Command: show power supply status, Description: This step involves verifying that power supply redundancy is restored after replacing the faulty component. The show power supply status command will display the current status of the power supplies.

Post-Check Verification

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Command: show system internal memory, Expected Output: No memory parity errors detected, Description: This command verifies that memory parity errors are no longer detected after the resolution steps are completed.

Command: show interface status, Expected Output: Line card status: Up, Description: This command verifies that the line card is now operational after the replacement step.

Command: show power supply status, Expected Output: Power supply redundancy: Restored, Description: This command verifies that power supply redundancy is restored after the replacement step.
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Roll-back Procedures

- Step: Revert to previous configuration, Command: load merge , Description: This step involves reverting to the previous switch configuration in case the resolution steps fail to resolve the issue. The load merge command will merge the previous configuration with the current one.

- Step: Restore power supply redundancy, Command: show power supply status, Description: This step involves restoring power supply redundancy in case the replacement step fails. The show power supply status command will display the current status of the power supplies.